

ANDREW MAKAREVICH

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EDUCATION

University of Maryland, College Park
B.S. Computer Engineering, GPA: 3.66/4.00

College Park, MD
Jan 2025 – Dec 2028

EXPERIENCE

Avionics Engineer — UMD Satellite Development Program
College Park, MD

Jan 2025 – Present

- Develop and test a Python-based **control and monitoring software system** for embedded hardware, supporting data acquisition, command execution, and real-time visualization.
- Design and validate **packet-based communication interfaces** with defined timing, framing, and data integrity checks to ensure reliable system behavior.
- Author and maintain **interface definitions and command mappings**, supporting integration testing, traceability, and operational documentation.
- Build **test harnesses and simulation tools** to emulate hardware behavior, enabling verification, fault injection, and repeatable debugging prior to deployment.

Research Assistant — Medical AR Lab
Johns Hopkins University

Jun 2023 – Aug 2023
Baltimore, MD

- Processed surgical video data in Python to support **real-time image segmentation** for augmented-reality surgical visualization systems.
- Integrated 3D anatomical assets into an AR pipeline to improve **registration accuracy and overlay precision** between physical and virtual anatomy.

Robotics Researcher — EXACT Robotics Lab
College Park, MD

Sep 2025 – Present

- Develop ROS2-based control frameworks for robotic systems using Python, emphasizing modularity and real-time performance.
- Integrate sensing, planning, and actuation modules in closed-loop control systems, with focus on robustness and predictable behavior.

Engineering Intern
Harbor Designs and Manufacturing LLC

Apr 2024 – Jun 2024
Baltimore, MD

- Created UI wireframes and early-stage prototypes to support engineering product development and stakeholder review.
- Collaborated across software and hardware teams to improve system usability, integration, and design clarity.

TECHNICAL PROJECTS

Interrupt-Driven Signal Detection System — STM32, Real-Time DSP

2025

- Developed FIR-based interrupt signal detection for low-latency system response; optimized ADC sampling and ISR timing.

TECHNICAL SKILLS

Programming: Python, Java, MATLAB, Embedded C

Systems: Embedded systems, real-time software, control and monitoring systems, data acquisition pipelines, test automation, packet-based communication

Tools: ROS2, Git, STM32CubeIDE, MATLAB, VS Code, SolidWorks, Onshape

Certifications: FCC Amateur Radio License (Technician)

Languages: English (Native), Russian (Native), French (Fluent), Spanish (Fluent), Mandarin (HSK 2)